

F.5 — Planar VTOL: Transfer-function model (about hover)

Use small-angle hover linearization with perturbations $\tilde{h}, \tilde{z}, \tilde{\theta}, \tilde{F}, \tilde{\tau}$. Let $F_e = mg$.

Vertical (longitudinal) channel:

$$m \ddot{\tilde{h}} = \tilde{F} \quad \Rightarrow \quad \frac{\tilde{H}(s)}{\tilde{F}(s)} = \frac{1}{m s^2}.$$

Pitch (attitude):

$$J \ddot{\tilde{\theta}} = \tilde{\tau} \quad \Rightarrow \quad \frac{\tilde{\Theta}(s)}{\tilde{\tau}(s)} = \frac{1}{J s^2}.$$

Lateral (side translation):

$$m \ddot{\tilde{z}} + \mu \dot{\tilde{z}} = -F_e \tilde{\theta}, \quad F_e = mg,$$

so

$$\frac{\tilde{Z}(s)}{\tilde{\Theta}(s)} = -\frac{F_e/m}{s^2 + (\mu/m)s}, \quad \frac{\tilde{Z}(s)}{\tilde{\tau}(s)} = \left(-\frac{F_e}{m} \right) \frac{1}{J s^2} \cdot \frac{1}{s^2 + (\mu/m)s}.$$

Block-diagram sketch (text).

- Vertical: single block $1/(ms^2)$ from $\tilde{F}$ to $\tilde{h}$.
- Lateral cascade: $\tilde{\tau} \rightarrow \tilde{\theta}$ via $1/(Js^2)$, then $\tilde{\theta} \rightarrow \tilde{z}$ via $-\frac{F_e}{m} \cdot \frac{1}{s^2 + (\mu/m)s}$.

F.6 — Planar VTOL: Linear state-space model

Choose

$$x = \begin{bmatrix} \tilde{z} & \tilde{h} & \tilde{\theta} & \dot{\tilde{z}} & \dot{\tilde{h}} & \dot{\tilde{\theta}} \end{bmatrix}^\top, \quad u = \begin{bmatrix} \tilde{F} \\ \tilde{\tau} \end{bmatrix}, \quad y = \begin{bmatrix} \tilde{z} \\ \tilde{h} \\ \tilde{\theta} \end{bmatrix}.$$

Linearized dynamics about hover (with lateral drag μ):

$$\ddot{\tilde{h}} = \frac{1}{m} \tilde{F}, \quad \ddot{\tilde{\theta}} = \frac{1}{J} \tilde{\tau}, \quad \ddot{\tilde{z}} = -\frac{\mu}{m} \dot{\tilde{z}} - \frac{F_e}{m} \tilde{\theta}, \quad F_e = mg.$$

State-space form $\dot{x} = Ax + Bu, y = Cx + Du$:

$$A = \begin{bmatrix} 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & -\frac{F_c}{m} & -\frac{\mu}{m} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}, \quad B = \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ \frac{1}{m} & 0 \\ 0 & \frac{1}{J} \end{bmatrix},$$

$$C = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \end{bmatrix}, \quad D = \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 0 \end{bmatrix}.$$

HW3 Testing — F Section (VTOL)

This mirrors your D-section test notes, but for **F.5 (transfer functions)** and **F.6 (linear state space)** using the hover-linearized VTOL.

1) VTOL dynamics unit tests (nonlinear `Dynamics.f(x,u)`)

Run

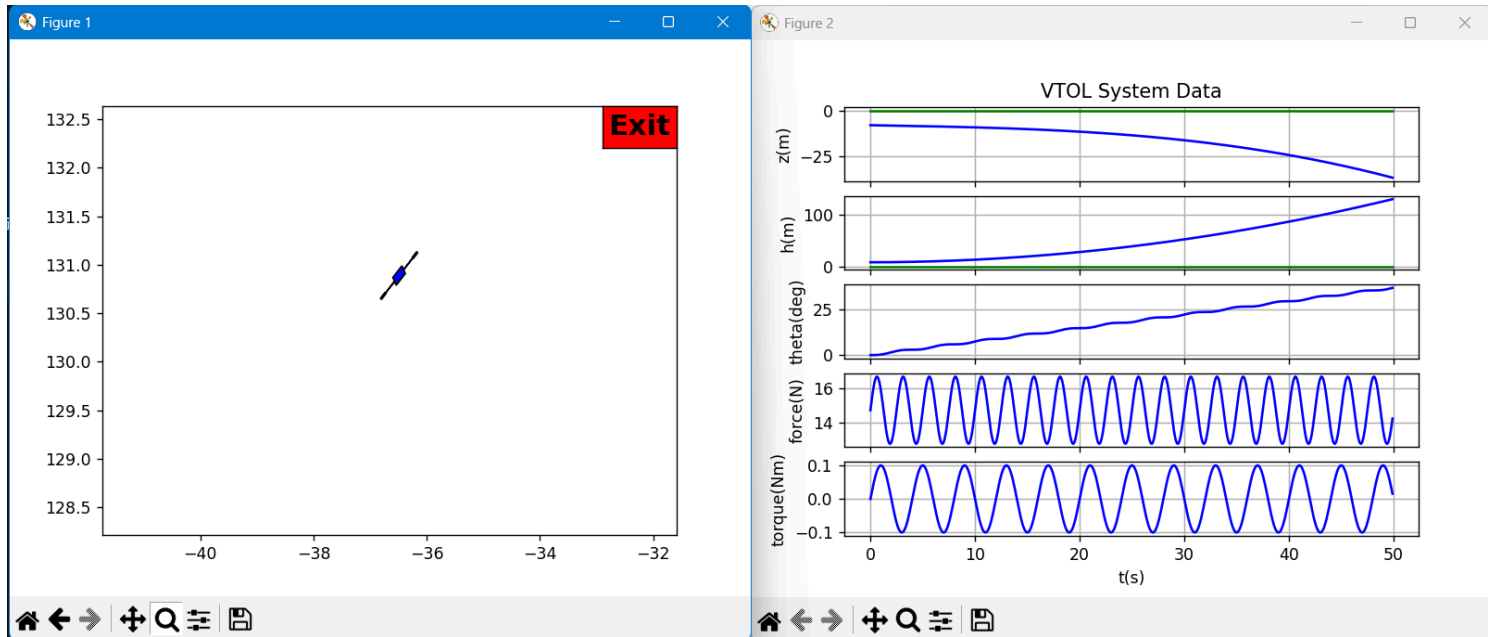
```
C:/Users/consa/Downloads/Programming/Robotics_Controls/.venv/Scripts/python.exe ^
c:/Users/consa/Downloads/Programming/Robotics_Controls/_F_planar_vtol/python/testDynamics.py
```

Expect

- The script executes without errors and reports each derivative check as **PASS** (exact text may differ from the mass test).
- This validates your **nonlinear** `Dynamics.f(x,u)` ODE (rotor inputs `[f_r, f_l]`) still behaves.

Tip: If you previously saw `AttributeError: VTOLParam has no attribute 'unmixing'` , make sure `VTOLParam.py` defines `mixing` (used to map `[F; τ] → [f_r; f_l]`). The linear demo uses `P.mixing` .

2) Linear hover model + animation (runVTOLAnimation.py)



Run

```
C:/Users/consa/Downloads/Programming/Robotics_Controls/.venv/Scripts/python.exe ^
c:/Users/consa/Downloads/Programming/Robotics_Controls/_F_planar_vtol/python/runVTOLAnimation.py
```

What the program does

- Simulates the **F.6 linear state-space